

Gray Code to 7-segment Display ~~Convert~~ Decoder .

Date : _____

Truth table :

	G_2	G_1	G_0	a	b	c	d	e	f	g
0 →	0	0	0	1	1	1	1	1	1	0
1 →	0	0	1	0	1	1	0	0	0	0
2 →	0	1	1	1	1	0	1	1	0	1
3 →	0	1	0	1	1	1	1	0	0	1
4 →	1	1	0	0	1	1	0	0	1	1
5 →	1	1	1	1	0	1	1	0	1	1
6 →	1	0	1	1	0	1	1	1	1	1
7 →	1	0	0	1	1	1	0	0	0	0

for a :

	(00) $\bar{G}_1 \bar{G}_0$	(01) $\bar{G}_1 G_0$	(11) $G_1 G_0$	(10) $G_1 \bar{G}_0$
(0) $\bar{G}_2$	1		1	1
(1) G_2	1	1	1	

$$a = \bar{G}_1 \bar{G}_0 + G_1 \bar{G}_2 + G_0 G_2$$

for b :

	(00) $\bar{G}_1 \bar{G}_0$	(01) $\bar{G}_1 G_0$	(11) $G_1 G_0$	(10) $G_1 \bar{G}_0$
(0) $\bar{G}_2$	1	1	1	1
(1) G_2	1			1

$$b = \bar{G}_2 + \bar{G}_0 G_2$$

for c :

	(00) $\bar{G}_1 \bar{G}_0$	(01) $\bar{G}_1 G_0$	(11) $G_1 G_0$	(10) $G_1 \bar{G}_0$
(0) $\bar{G}_2$	1	1		1
(1) G_2	1	1	1	1

$$c = \bar{G}_1 + \bar{G}_0 + G_1 G_2$$

for d :

	(00) $\bar{G}_1 \bar{G}_0$	(01) $\bar{G}_1 G_0$	(11) $G_1 G_0$	(10) $G_1 \bar{G}_0$
(0) $\bar{G}_2$	1		1	1
(1) G_2		1	1	

$$d = G_0 G_1 + G_0 G_2 + \bar{G}_0 \bar{G}_2$$

for e :

	(00) $\bar{G}_1 \bar{G}_0$	(01) $\bar{G}_1 G_0$	(11) $G_1 G_0$	(10) $G_1 \bar{G}_0$
(0) $\bar{G}_2$	1		1	
(1) G_2		1		

$$e = \bar{G}_0 \bar{G}_1 \bar{G}_2 + \bar{G}_1 G_0 G_2 + G_1 G_0 \bar{G}_2$$

for f :

	(00) $\bar{G}_1 \bar{G}_0$	(01) $\bar{G}_1 G_0$	(11) $G_1 G_0$	(10) $G_1 \bar{G}_0$
(0) $\bar{G}_2$	1			
(1) G_2		1	1	1

$$f = \bar{G}_2 \bar{G}_1 \bar{G}_0 + G_0 \cdot G_2 + \cancel{G_1 G_2} G_1 G_2$$

for g :

	(00) $\bar{G}_1 \bar{G}_0$	(01) $\bar{G}_1 G_0$	(11) $G_1 G_0$	(10) $G_1 \bar{G}_0$
(0) $\bar{G}_2$			1	1
(1) G_2		1	1	1

$$g = G_1 + G_0 \cdot G_2$$